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Characteristics of Day-roosts Used by Northern Long-eared Bats (*Myotis septentrionalis*) in Coastal New York

Katherine M. Gorman^{1,*}, Elaine L. Barr², Tomás Nocera³, and W. Mark Ford⁴

Abstract - In North America, *Myotis septentrionalis* (Northern Long-eared Bat) has experienced precipitous declines from white-nose syndrome. As these bats become rare and difficult to capture, additional day-roost assessments to inform management may fill gaps in our understanding, particularly in habitats and regions where such roosts have never been surveyed. Over 2 summers, we radio-tracked 16 individuals from a maternity colony on Long Island, NY, in a small forested patch surrounded by development and ocean. These bats disproportionately selected small, suppressed *Robinia pseudoacacia* (Black Locust) trees or snags for roosting. Generally, roosts occurred within the interior or edges of this forest patch, rather than surrounding suburbia, reinforcing the hypothesis that Northern Long-eared Bats are forest adapted. Our study shows even small tracts of forest in coastal, urban areas may have conservation value in providing day-roost and foraging habitat.

Introduction

As global diversity of bat species declines due to anthropogenically induced stressors such as habitat loss, degradation, and fragmentation, particularly through urbanization (Thalke et al. 2018), the forest-obligate *Myotis septentrionalis* (Trouessart) (Northern Long-eared Bat) faces an additional threat posed by the fungal pathogen *Pseudogymnoascus destructans* (Blehert & Gargas) Minnis & D.L. Lindner that causes white-nose syndrome (WNS). Since the advent of WNS in North America, the population of Northern Long-eared Bats has declined an estimated >90% throughout 79% of their range (Cheng et al. 2021). Accordingly, in 2012, the Canadian government declared Northern Long-eared Bats as endangered (Forbes 2012), and subsequently, the species was listed as threatened in New York (NYDEC 2019) and in the United States (USFWS 2015). Management actions focused on improving, increasing, or generally providing roosting habitat are vital to the survival of remaining colonies on the landscape (Silvis et al. 2016).

Following hibernation, reproductive female Northern Long-eared Bats typically return to the same general areas on the landscape to form maternity colonies and raise their young (Johnson et al. 2012, Perry 2011, Silvis et al. 2014). Due to their ability to maneuver easily in dense vegetation and glean prey directly from foliage

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(Faure et al. 1993, Silvis et al. 2016), this species often roosts and forages in mature forest or interior canopy gaps (Bergeson et al. 2021, Divoll et al. 2022, Owen et al. 2002). In most day-roosting (hereafter roosting) studies, Northern Long-eared Bats preferred cavities to exfoliating bark and snags in increasing stages of decomposition over live trees (Broders and Forbes 2004, Carter and Feldhamer 2005, Grider et al. 2021, Menzel et al. 2002, Rojas et al. 2017, Sasse and Pekins 1996, Silvis et al. 2012). However, local roost preferences are influenced by availability of tree species and forest condition (Perry and Thill 2007), as well as phase in the female reproductive cycle, which presumably influences selection based on thermoregulatory needs (Garroway and Broders 2008, Drake et al. 2020, Patriquin et al. 2010, Thalkin and Lacki 2018).

Most studies of roosting by Northern Long-eared Bats have occurred in large, forested landscapes (e.g., Broders and Forbes 2004, Ford et al. 2016, Johnson et al. 2012, Kalen et al. 2022, Menzel et al. 2002, Perry and Thill 2007, Silvis et al. 2012). Despite adaptations to forested environments, several observations of Northern Long-eared Bats have occurred post-WNS in fragmented or developed landscapes such as Martha's Vineyard, MA (Dowling and O'Dell 2018); Washington, DC (Deeley 2019); and Toronto, ON, Canada (Thorne et al. 2021); as well as in coastal areas formerly believed outside their range (Dowling and O'Dell 2018, Jordan 2020, Morris et al. 2009). Individual Northern Long-eared Bats from Long Island, NY, were initially identified and banded in 2015 (Lindsay Ries, US National Park Service, Fire Island National Seashore, NY, 4 March 2019 pers. comm.), and a maternity colony was documented in 2018 (Nocera 2018), presenting an opportunity to study roosting preferences of a coastal population in a comparatively urban landscape.

Habitat conditions and year-to-year variation in weather influence roost selection by Northern Long-eared Bats (Patriquin et al. 2016, Perry and Thill 2007), and examining multi-year roost characteristics across a range of geographic localities is important to inform future regional and localized conservation of the species. Our objective for this study was to characterize maternity roosts and roosting area over 2 years in a small, fragmented forest surrounded by anthropogenic development and the ocean, which is unlike most previous settings in which research has occurred on day-roosts of Northern Long-eared Bats (Gorman et al. 2022, Silvis et al. 2016).

Northern Long-eared Bats apparently prefer certain tree species for roosting over others, although selection varies with composition of the local forest (Silvis et al. 2016). For example, in the central Appalachians and Ohio River Valley, if the forest community does not include favored species in abundance, e.g., *Robinia pseudoacacia* L. (Black Locust; Ford et al. 2021, Johnson et al. 2009), Northern Long-eared Bats gravitate to other tree species with similar structural forms and longevity, e.g., *Sassafras albidum* (Nutt.) Nees (Sassafras; Silvis et al. 2012). If forest composition and structure is less than ideal, these bats may demonstrate even greater plasticity in roost selection on a day-to-day basis, including use of anthropogenic structures (De La Cruz et al. 2018). We hypothesized that Northern Long-eared Bats on Long Island, similar to those in some other regions, would roost

in cavities of Black Locusts or snags of similar long-standing species (Ford et al. 2006). In addition, we predicted that roosting would occur close to water or other commuting corridors (Divoll et al. 2022, Gorman et al. 2022, Rojas et al. 2017) and within interior forest, rather than in anthropogenic structures in surrounding neighborhoods or forest edges (Thorne et al. 2021). However, we also anticipated some differences in roost selection, relative to previous studies, due to limited availability of older, mature forest on Long Island.

Field-site Description

We conducted our study on Long Island, NY, at the 248-ha William Floyd Estate (hereafter, the estate) that is part of the Fire Island National Seashore, a unit of the US National Park Service (Fig. 1). The Atlantic Ocean borders the estate to the south, and the rest is surrounded by urban/suburban neighborhoods. The estate is primarily composed of mature, unmanaged northern coastal *Quercus* (Oak)–Eriaceae Heath forest (65% of total land cover; Gorman et al. 2021, Klopfer et al. 2002). Common tree species are *Pinus rigida* Mill. (Pitch Pine), *Q. alba* L. (White Oak), *Q. coccinea* Münchh. (Scarlet Oak), Sassafras, *Prunus serotina* Ehrh. (Black Cherry), *Acer rubrum* L. (Red Maple), Black Locust, and *Nyssa sylvatica* Marshall (Blackgum). The remainder of the estate consists largely of tidal salt marshes (25%), roads/trails, buildings, and open fields or other areas of sparse vegetation (10%). Average rainfall from May to July is ~10.2 cm; average low temperature during these months is 13.7 °C and varies from 8.9 °C in May to 17.8 °C in July, whereas average high temperature is 23.9 °C and varies from 19.4 °C in May to 27.8 °C in July (US Climate Data 2022). For a more-detailed description of the local climate, see Gorman et al. (2021).

Methods

Mist-netting and tracking

We conducted mist-netting from 16 May to 26 July 2018 and 15 March to 30 July 2019. We placed mist-netting systems that were 4.8–7.2 m tall and 4–12 m wide (Avinet Research Supplies, Portland, ME) over linear landscape features, such as roads and trails (Deeley 2019, Silvis et al. 2012). We recorded demographic data including species, sex, age class (adult or juvenile; Brunet-Rossinni and Wilkinson 2009), reproductive status, and body mass (g) for all captured bats. Upon first capture, we fitted all Northern Long-eared Bats with a uniquely numbered 2.9-mm aluminum alloy band (Porzana, East Sussex, UK) for subsequent identification. We affixed radio-transmitters (0.27 g; model LB-2X; Holohil Systems, Carp, ON, Canada), centered between the scapulae, using a surgical cement (Perma-Type, Plainville, CT). All adult females that were radiotracked weighed at least 5.4 g, so transmitters were ≤5% of body mass (Aldridge and Brigham 1988), and a single juvenile male that received a transmitter weighed 6.2 g. All handling of bats followed protocols approved by the Institutional Animal Care and Use Committee of the Virginia Polytechnic Institute and State University,

and field work was conducted under the authority of permits issued by the New York Department of Environmental Conservation, US National Park Service, and US Fish and Wildlife Service. We subsequently tracked these bats to roosts using TRX-1000S receivers and matched 3-element Yagi antennas (Wildlife Materials,

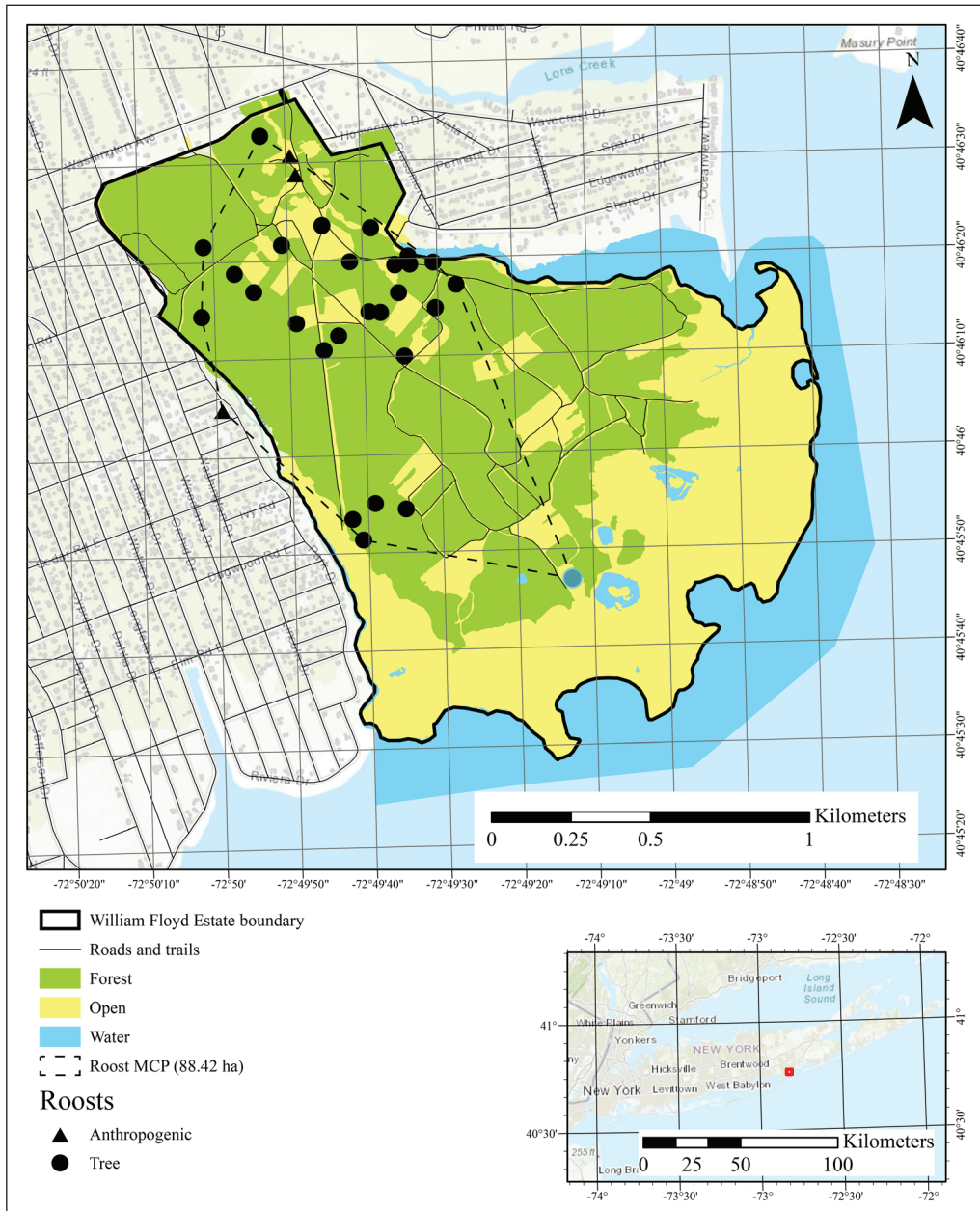


Figure 1. Map of roosts used by Northern Long-eared Bats at the William Floyd Estate, NY (2018–2019). Roost MCP is a minimum convex polygon defining minimum roosting area over both years combined. Red rectangle on the inset map (ESRI, Redlands, CA) indicates study area.

Carbondale, IL) every day for the life of the transmitter, until it fell off or until the bat was not located for 3 consecutive days. We georeferenced roosts with handheld GPS units (Garmin International, Olathe, KS) and plotted all coordinates in ArcPRO (Version 2.5.0, ESRI, Redlands, CA).

Roost characteristics

We tracked bats during the day, and confirmed occupancy of trees by performing emergence counts every night that a bat with a transmitter was in a roost, weather permitting. During the day following discovery of a new roost, we took measurements of the roost tree and the 4 closest nonroost trees having a diameter at breast height (DBH) > 10 cm as “pseudo-absences”; these were not necessarily true absences, because actual bat use was unknown, but they represented trees that were available to the bats on the landscape (Barbet-Massin et al. 2012, Grider et al. 2021). One nonroost tree was selected in each cardinal direction using the point-quarter method (Jorge et al. 2021). We included species that have not been historically reported as roosts for Northern Long-eared Bats because we questioned whether there were inherent interspecific structural differences among tree species or if Northern Long-eared Bats roost in unlikely species in emergency situations. We used a clinometer (model PM-5/1520; Vantaa, Finland) to measure heights of the roost tree and the emergence points, as well as the height of nonroosts. We measured diameter at breast height with a DBH tape and distance from the roost to each nonroost tree with a tape measure. For each roost and nonroost, we identified species, crown class (1–4: dominant, codominant, intermediate, or suppressed, respectively; Nyland 1996), and decay stage (1–7: alive, declining, dead, loose bark, clean, broken top, or stump, respectively; Cline et al. 1980). We confirmed whether the emergence point was a cavity or loose bark, recorded the percent bark remaining on the roost, measured the basal area with a 20-factor prism (Forestry Suppliers, Jackson, MS; Thompson et al. 2006), and determined canopy closure using a spherical concave densiometer (Forestry Suppliers, Jackson, MS; Russavage et al. 2021). All means are stated as ± 1 SD.

Statistical analyses

We performed all statistical tests and analyses using R version 4.0.3 (R Core Team 2020). To compare attributes of roosts versus nonroosts, we made a series of binomial generalized linear mixed models (GLMM; Brooks et al. 2017), with roost versus nonroost as the response variable, total height, DBH, crown class, and decay stage as the predictor variables, and individual roost as the random effect (random trees were assigned the closest roost tree). Prior to model design, we tested for collinearity among our predictor variables using Pearson’s correlation coefficient (Wei and Simko 2017) and found a strong correlation ($>|0.6|$) between only total height and DBH; thus, we did not include those 2 variables in the same models. We created a set of 8 a priori candidate models, including a null model and 2 global models (1 with total height and 1 with DBH). The remaining models were based on our knowledge of the species’ ecology and combined structural characteristics (height and crown class, indicating perhaps the amount of solar exposure) with

conditional ones (decay stage and DBH, which may influence the internal temperature). We then ranked each model using Akaike's information criteria (AIC) and considered all models within $\Delta AIC < 2$ to be potentially informative (Burnham and Anderson 2002). Further, we examined model weights and ecological soundness of top-performing models. We checked each model for goodness-of-fit and over- and under-dispersion in the form of a QQ-plot, residual plot, and a one-sample Kolmogorov–Smirnov test (Hartig 2020). We evaluated the strength of a predictor variable with a confusion matrix, area under the curve (AUC), and a receiver-operating-characteristic graph (Sing et al. 2005). The GLMM served as a broad preliminary exploration of differences between trees used as roosts compared to nearby trees that, in theory, were not selected by Northern Long-eared Bats.

To determine if a tree species was used for roosting more or less often than expected based on availability, we performed a Pearson's chi-square test (Venables and Ripley 2002). We omitted 3 anthropogenic roosts, a bat house and 2 wooden utility poles from this analysis, because there were no other ones within the estate, and we also grouped any tree species with less than 5 observations into a category labelled "other" (Neu et al. 1974). We used $\alpha = 0.05$ as our threshold to determine the significance of the test and used residuals and expected values to determine the relationships between Northern Long-eared Bats and species of trees.

To evaluate landscape characteristics, we collected several spatial attributes relative to roosting locations. We measured the distance from roosts to various types of land cover with the 'Near' tool in ArcGIS Pro. We determined the location of forest and open areas (classes: developed and open space), using the National Land Cover Dataset (Multi-Resolution Land Characteristics Consortium 2016); water bodies, using the US Geological Survey National Hydrography Dataset (USGS 2020); and roads, using centerline data for Suffolk County, NY (Suffolk County Open Data 2020), and a roads and trails layer provided by the US National Park Service.

After combining these spatial data with the characteristics of the roost trees (total height of roost, height of bat emergence, DBH, crown class, and decay stage), we performed a principal components analysis (PCA; Oksanen et al. 2020), to assess the whole of roost characteristics from a multivariate approach. We chose PCA as an additional step in our overall analysis to combine numerous, often correlated factors into an interpretable series of wholly orthogonal or independent factors. We selected the first 2 principal components for visualization (Wickham 2016), based on examination of proportion of variance explained, eigenvalues, and the screeplot.

Results

In 2018, we captured and affixed transmitters to 14 adult female Northern Long-eared Bats (1 of which was banded as an adult in 2015) and tracked them to 21 unique roosts (19 trees, 1 bat house, and 1 utility pole). In 2019, we captured and affixed transmitters to 3 adult females (2 of which were banded and tracked as adults in 2018) and 1 juvenile male. These bats were tracked to 10 unique roosts: 1 of which was a wooden utility pole that also was used in 2018, an additional wood-

en utility pole, and 8 trees. Across both years, 16 unique individuals were located on 1–10 days each, for an average of 4 ± 2 (SD) days, and followed to 30 unique roosts. In addition to the single juvenile, 8 radiotracked adults were pregnant, 7 were of unknown reproductive status (not obviously pregnant, though potentially within early pregnancy), and 1 was lactating (unknown reproductive status in 2018, but lactating in 2019). Given that reproductive status may have changed during the tracking periods, coupled with the large number of ambiguous reproductive states, we decided to omit reproductive status as a variable in our analysis. After combining data from both years, we calculated a minimum roosting area for the maternity colony of 88.4 ha (Fig. 1).

Roosts included 3 anthropogenic structures (2 wooden utility poles and one 3-chambered traditional bat house) and 10 species of trees, predominantly Black Locust ($n = 14$; see Table S1, available online in Supplemental File S1 at <http://www.eaglehill.us/NENOnline/suppl-files/n29-2-N1961-Gorman-s1>, and for BioOne subscribers, at <https://www.doi.org/10.1656/N1961.s1>). Nonroost trees were predominantly Black Locust ($n = 22$) and Red Maple ($n = 19$) (see Table S1 in Supplemental File S1). For roosts, total height (including anthropogenic structures) was 9.0 ± 4.9 m, roosting height was 4.1 ± 2.0 m, and DBH was 22.5 ± 14.0 cm (Fig. 2). For nonroost trees, total height was 9.5 ± 7.2 m, DBH was 20.7 ± 12.8 cm (Fig. 2), and distance to the roost tree was 5.4 ± 5.4 m. Crown classes of roosts trees were mostly codominant or suppressed (34% and 41%, respectively) and evenly distributed across all decay stages, with the exclusion of stump (see Table S2 in Supplemental File S1). Most nonroost trees were alive (64%), and approximately one-third (36%) were suppressed. Roosts were located 44.0 ± 34.3 m from the

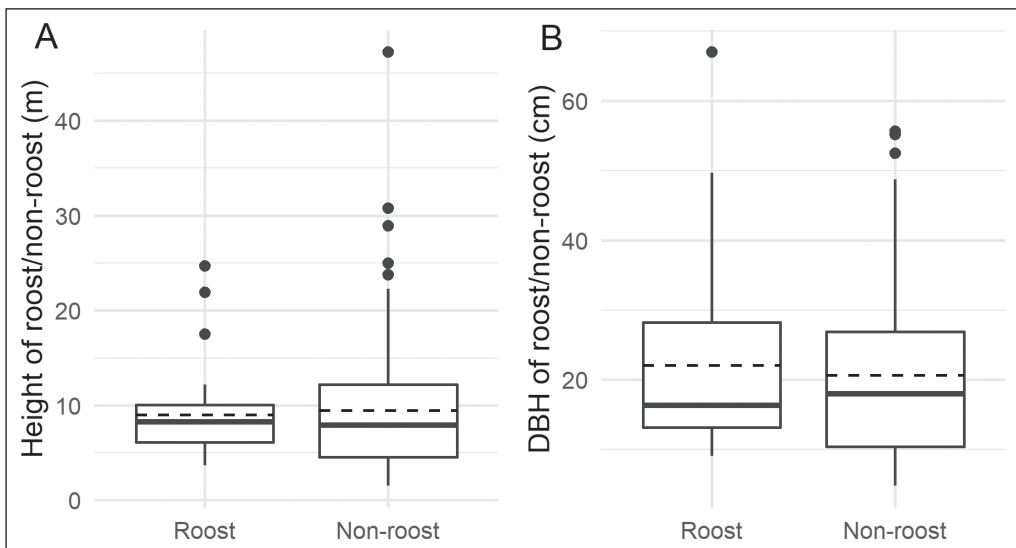


Figure 2. Boxplots of height (A) and diameter at breast height (DBH; B) for summary data of roosts used by Northern Long-eared Bats, including anthropogenic structures, and non-roost trees. Dashed lines represent means, dots represent outliers, tails represent 95th percentile, box bounds represent 50th percentile, and solid lines in boxes represent median values.

nearest road, 268.8 ± 185.4 m from fresh water or tidal streams, 4.7 ± 14.2 m from forest (23 roosts within forest, 6 outside forest), and 25.3 ± 30.6 m from the nearest open field (Fig. 3).

Our top-performing GLMM had a model weight of 0.73 and included decay stage and crown class individually and as an interactive term, with no other model coming within 2 Δ AIC units (Table 1). Alone, crown class was not a significant

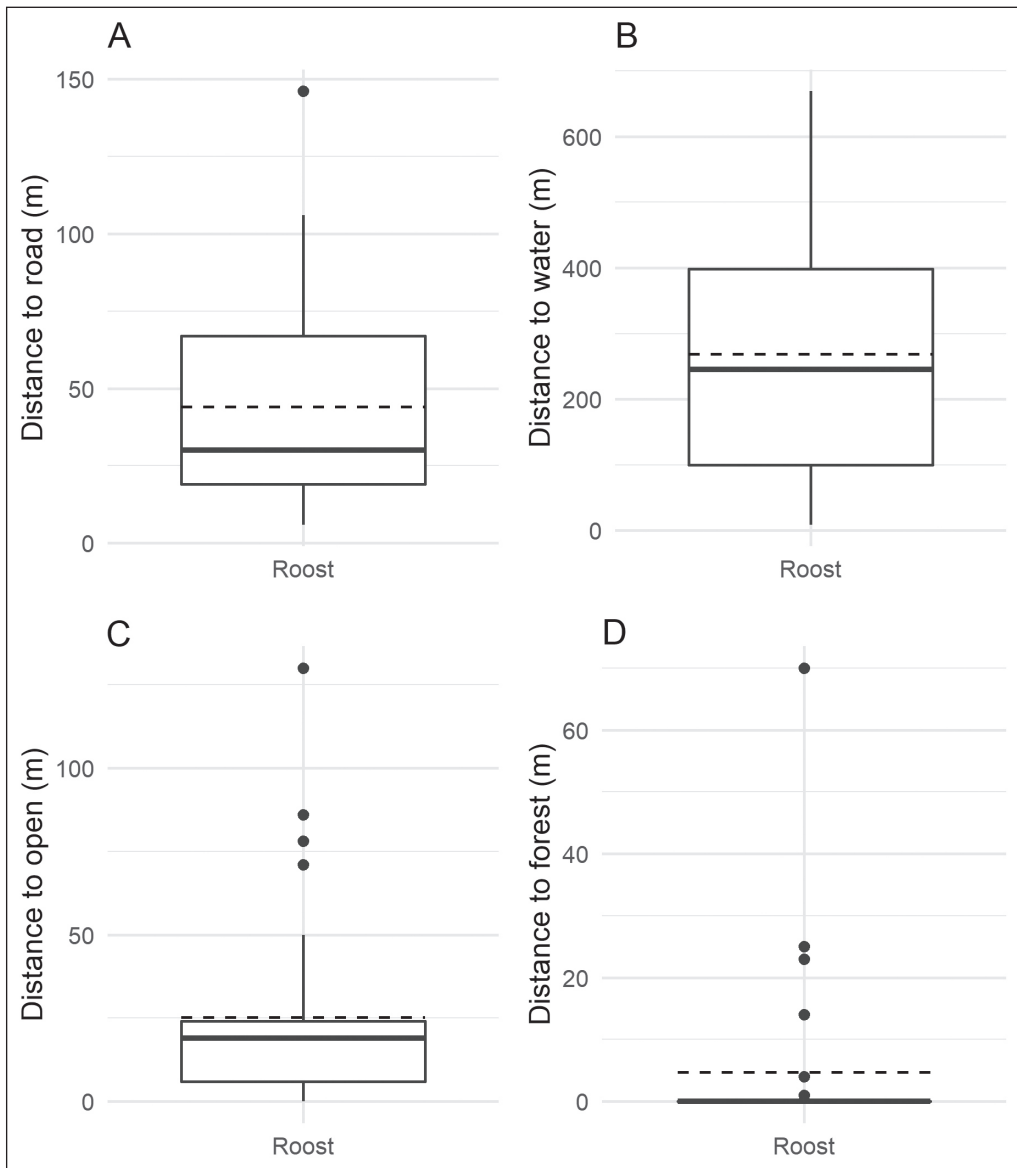


Figure 3. Boxplots of distance to road (A), water (B), open (C), and forest (D) for roosts used by Northern Long-eared Bats, including anthropogenic structures. Dashed lines represent means, dots represent outliers, tails represent 95th percentile, box bounds represent 50th percentile, and solid lines in boxes represent median values. Note the varying y-axes values.

predictor in roost selection (confidence intervals crossed zero), whereas decay stage alone had a statistically significant positive relationship with roost selection (confidence intervals did not cross zero), but for the 2 variables combined, there was a negative relationship with roost selection (confidence intervals did not cross zero) (Table 2).

Our Pearson’s chi-square revealed a relationship between roost species and use by Northern Long-eared Bats ($\chi^2_8 = 18.3$; $n = 132$; $P = 0.02$; Table 3). Unidentified snags and Black Locusts were used significantly more often than expected based on availability, and all other species were used equitably or slightly less often than expected (Table 3).

From our PCA, the first 3 components combined for 69.4% of variance in roost characteristics, with the first, second, and third components explaining 35.5%, 20.5%, and 13.4%, respectively (see Table S3 in Supplemental File 1). The first component was most representative of general roost characteristics and had high loadings of total roost height (1.10), DBH (1.09), and distance to water (-0.97). The second component was primarily representative of decay stage (-0.96) and crown class (-0.86), whereas the third component represented distance to road (0.86) and basal area (-0.70). Our PCA identified clear clustering in roost characteristics among Black Locusts and snags, with Red Maple often falling on the opposite side of the graph, meaning Black Locust was favorable for 1 combination of variables, whereas Red Maple was favored under opposite circumstances (these 2 trees were the species most commonly used as roosts; Fig. 4). There was a strong association

Table 1. Akaike’s information criteria (AIC) table showing all models testing relationship of roost selection by Northern Long-eared Bats at the William Floyd Estate, NY (2018–2019) to roost variables, AIC score, delta AIC, and model weights (ω_i); “*” denotes an interaction. Predictor variables included crown class, decay stage, height of potential or actual roost in meters (height), and diameter at breast height (DBH) of potential or actual roost. Roost ID was included in all models as a random effect.

Model	AIC	Δ AIC	ω_i
Crown class * decay stage	128.75	0.00	0.73
Height * decay stage	133.09	4.34	0.08
DBH + crown class + decay stage	133.32	4.56	0.07
Height + crown class + decay stage	133.58	4.82	0.07
Height + decay stage	134.60	5.85	0.04
Null model	138.97	10.22	<0.01
Height + crown class	142.48	13.73	<0.01
DBH + crown class	142.76	14.01	<0.01

Table 2. Predictor variables, beta (β) values, standard errors (SE), lower (LCL) and upper (UCL) 95% confidence intervals for top model output of roost selection by Northern Long-eared Bats at the William Floyd Estate, NY (2018–2019) related to roost variables; “*” denotes an interaction.

Predictor variables	β	SE	LCL	UCL
Crown class	0.21	0.38	-0.54	0.97
Decay stage	1.35	0.49	0.38	2.32
Crown class * decay stage	-0.28	0.14	-0.56	-0.01

Table 3. Standardized Pearson’s residuals and expected values for chi-squared analysis, comparing species of nonroost and day-roost trees used by Northern Long-eared Bats at the William Floyd Estate, NY (2018–2019). Residual values close to |2.00| are considered influential (indicated by “†”; Silvis et al. 2015b). Negative and positive values indicate lesser and greater abundance than expected, respectively. “Other” includes: *Picea abies* (L.) Karst (Norway Spruce), *Juniperus virginiana* L. (Eastern Redcedar), White Oak, *Q. ilicifolia* Wangenh (Bear Oak), *Ulmus americana* L. (American Elm), Sassafras, and *Diospyros virginiana* L. (Eastern Persimmon).

Tree species	Residuals		Expected		Use vs. Availability
	Roost	Random	Roost	Random	
<i>Carya</i> spp. Nutt. (Hickory)	-0.69	0.34	1.97	8.03	Lesser
<i>Quercus rubra</i> L. (Northern Red Oak)	-0.89	0.44	2.36	9.64	Lesser
<i>Lindera benzoin</i> (L.) Blume (Spicebush)	-1.09	0.54	1.18	4.82	Lesser
Black Cherry	-1.54	0.76	2.36	9.64	Lesser
Black Locust	2.33†	-1.15	6.89	28.11	Greater
Red Maple	-0.64	0.32	4.33	17.67	Lesser
Blackgum	-0.24	0.12	2.36	9.64	Lesser
Other	-0.56	0.28	2.95	12.05	Lesser
Unidentified snag	1.93†	-0.96	1.58	6.42	Greater
χ^2	18.33				
P-value	0.019				

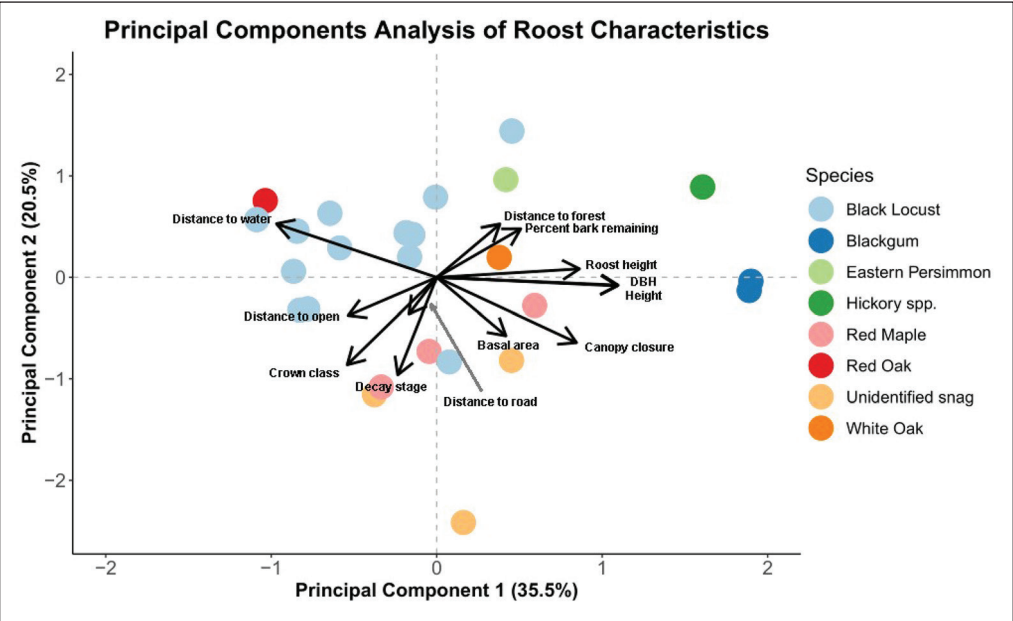


Figure 4. Principal components 1 and 2 for roosts used by Northern Long-eared Bats at the William Floyd Estate, NY (2018–2019). Arrow lengths indicate effect size. Direction of arrow relative to species points indicates positive or negative relationship. For example, in this figure, there is a positive relationship with Black Locust as distance to water increases. There is a negative relationship with Black Locust as canopy closure and height increase. Diameter at breast height (DBH) and height arrows overlap. See text for scientific names of tree species in figure legend.

with use of short, low-DBH Black Locust with increased distance to water, which was the opposite for Blackgum and, to a lesser extent, Red Maple. As the basal area (a surrogate for clutter or stem density) increased, there was a strong association of unidentified snag use.

Discussion

A roosting area of 88.4 ha for the maternity colony on Long Island was more than that observed in studies in Canada, West Virginia, Kentucky, and Wisconsin (min–max = 0.39–58.5 ha; Henderson and Broders 2008, Johnson et al. 2012, Silvis et al. 2015a, Hyzy et al. 2020). Smaller roosting areas could reflect short tracking periods (i.e., if bats were not followed long enough to identify the entire scope of their inter-roost movements) or result from “condensed” but superior habitat in terms of roost availability and foraging conditions (Ford et al. 2016; Johnson et al. 2009, 2012). In our study, despite tracking periods on par with other studies, the larger roosting area may reflect poor roosting and foraging habitat within and around the estate.

Although other studies have shown that Northern Long-eared Bats occasionally roost in anthropogenic structures, those structures typically were located within or on the edge of forested habitat (De La Cruz et al. 2018, Dowling and O’Dell 2008, Henderson and Broders 2008). Despite the abundance of anthropogenic structures that could double as roosts in the surrounding urban area of Long Island (e.g., houses and wooden utility poles), we found no evidence to suggest that the bats of this colony moved far from the forests at the estate; the bat house was the only roost outside the forest, but it was located 79 m from the edge. The closest patches of woodland, other than trees associated with anthropogenic structures outside the estate, were >3 km beyond neighborhoods to the west in the Wertheim National Wildlife Refuge or >1 km to the south over open ocean on Fire Island. Although we cannot say definitively that members of this colony were limited to the estate, we did not note any movements to the available forests to the west or south during our tracking. Though variable with numerous instances to the contrary, most research has shown that the combined roosting and foraging range of female Northern Long-eared Bats is <300 ha (Broders et al. 2006, Johnson et al. 2012, Owen et al. 2003), further suggesting that movement to the west or south to Fire Island is unlikely, based on both distance and habitat unsuitability.

The colony’s use of the estate and avoidance of surrounding areas likely is related to several factors. First, there is less artificial light within the forest at the estate, and increases in light pollution typically decrease bat activity (Rowse et al. 2016) because of increased predation risk and decreased prey diversity and abundance (Hakami et al. 2020, Threlfall et al. 2012, Villarroya-Villalba et al. 2021). Second, constant vehicular traffic may be an additional deterrent (Thorne et al. 2021). The density of human development (and thus light pollution and traffic) surrounding the estate could serve to concentrate prey, forcing Northern Long-eared Bats to forage within the forest or along its edge. This hypothesis is supported by an acoustic analysis at the estate, which noted that activity of Northern Long-eared Bats was

highest within forested patches and edges (Gorman et al. 2022). The estate forms a tract of forested land that is geographically isolated from similar habitat, thus providing a terrestrial, “island” refugium, and potentially indicating that the area contains at least the minimum requirements for a successful maternity colony. The Atlantic Ocean serves as an additional natural barrier to movement and also creates highly variable weather conditions that may necessitate numerous roosts with differing structural characteristics and microclimates to accommodate changing physiological needs throughout the maternity season.

Height and DBH were comparable between roosts and random trees in our study, pointing to no superficial structural differences between the 2 groups other than roosts tended to be more advanced in the decay process than random trees. This finding is congruent with other studies of roosting by Northern Long-eared Bats that have shown a preference for snags versus live trees (Carter and Feldhamer 2005, Grider et al. 2021, Rojas et al. 2017, Silvis et al. 2012). Our mean DBH is comparable to the diameter of maternity roosts of Northern Long-eared Bats in coastal North Carolina (20.6 cm; Jordan 2020) but much less than the mean DBHs from studies at inland and montane sites (31.1–65.0 cm; Broders and Forbes 2004, Carter and Feldhamer 2005, Foster and Kurta 1999, Grider et al. 2021, Rojas et al. 2017, Silvis et al. 2015b). Moreover, because the relationship between selection and decay stage was reversed when crown class was added as an interactive term, we found evidence for selection of snags in lower crown classes, similar to Johnson et al. (2009). Our beta value was lower for the interactive term as compared to decay stage alone; therefore, it is likely that this relationship is an artifact of the selected roosts being evenly distributed across the crown classes (tempering the strong positive effects of decay stage). The PCA further highlighted use of shorter and thinner (low DBH) Black Locusts and snags in areas of high-clutter and shade, which is opposite the typical pattern of large tall trees with high solar exposure reported for other species (Barclay and Kurta 2007), but consistent with the GLMM results.

When we compared across tree species for roost use, the significant residuals further confirmed Black Locust and unidentified snags in use more often than expected based on their availability among our random sample of trees. The demonstrated longevity, hardness, and rot-resistance of Black Locust (Adams and Owens 2001, Cuno 1930, Ford et al. 2006, Meier 2022), coupled with its reputation for cavity formation (Carey 1983, McComb and Muller 1983), may make it the ideal roost for the species, as its presence may be dependable year after year, and there may be several cavity options on the bole itself. Other studies of maternity roosts of Northern Long-eared Bats reported a selection for Black Locust, particularly in areas where it was abundant (Ford et al. 2006, Hyzy et al. 2020, Johnson et al. 2009, Menzel et al. 2002, Owen et al. 2002). The clustering of Black Locust directly opposite other species on the PCA plots suggests that female Northern Long-eared Bats identify and may be selecting specific roosts that provide optimal thermoregulatory conditions (Garroway and Broders 2008, Patriquin et al. 2010, Silvis et al. 2015a, Thalkin and Lacki 2017). Within the context of our study, thermoregulatory

needs could have been dictated by reproductive status (Garroway and Broders 2008, Patriquin et al. 2010, Silvis et al. 2012), maritime-influenced weather (Gorman et al. 2021), some combination of these 2 variables, or other factors entirely.

Despite distance to water having a strong association with roosts, it seems that the association was largely species-based, with Black Locusts falling mostly on one side of the PCA and all other species scattered on the opposite end. This result could reflect the specific habitats where those tree species are found (some trees are more common in the marshy areas of the estate) and that Northern Long-eared Bats base their roost selection on environmental cues coupled with their thermoregulatory needs. Further research with larger sample sizes, more diversity in reproductive stages, and longer tracking periods are needed to explore the mechanisms behind these patterns and how they might be similar to or differ from better-studied inland populations.

Our research at the estate provides additional insights into the post-WNS roosting ecology of Northern Long-eared Bats in a coastal and urbanized setting. Much like Northern Long-eared Bats in an urban forest in Toronto (Thorne et al. 2021), the population at the estate made little use of the anthropogenic environment around them, further highlighting the importance of conserving heterogeneous forest patches for bats, even in areas of high human occupancy. Whereas some aspects of roost selection and use differed from previous research at inland sites, roosting ecology of Northern Long-eared Bats on Long Island generally conformed to that of inland locations. Given the post-WNS rarity of Northern Long-eared Bats throughout their distribution along with the paucity of roosting data from urban and coastal areas, the observations we made may be useful to inform species-specific conservation protocols going forward. Whether or not coastal populations, such as the one we studied, are unaffected by WNS and are successfully reproducing is unknown, but our findings may assist resource personnel with identifying and managing other areas in the coastal Northeast with potential suitable and perhaps occupied habitat. In addition, our study shows even small tracts of forest in coastal, urban areas may have conservation value in providing day-roost and foraging habitat.

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